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BMDS2114 Machine Learning

Assignment

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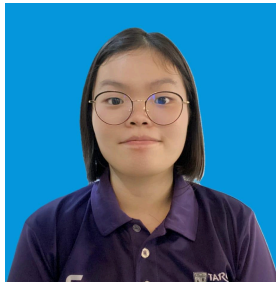
			
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PART A - Identifying and Selecting Promising Models

1.0 Introduction

Text clustering is an unsupervised machine learning technique that groups similar text documents into clusters, enabling the discovery of patterns and efficient organization of large text datasets without prior labeling (Aggarwal & Zhai, 2012). In the context of environmental news, it can be used to categorize articles by topics like climate change, pollution, conservation, and environmental policies, allowing media outlets, researchers, and policymakers to track trends, monitor topic shifts, and analyze public discourse effectively (Yin et al., 2012).

1.1 Problem Statement

In sustainability, the vast number of news articles across subtopics like climate change and pollution makes it difficult for experts to group related content efficiently before research. The explosion of unstructured digital content (Camacho-Collados & Pilehvar, 2018), fragmented information, and the time-consuming nature of manual methods (Jelodar et al., 2019) create major challenges. Without an organized structure, it is hard to track trends, compare regions, or support long-term sustainability analysis.

1.2 Objective of the Project

The main objective of this project is to apply text clustering techniques to a collection of environmental news articles to identify and group related topics or themes. Specifically, the project aims to:

- Implement unsupervised learning algorithms (e.g., Mini-Batch K-Means, Fuzzy C Means, Spectral Clustering and so on) on textual data.
- Preprocess text data through tokenization, stopwords removal, TF-IDF vectorization and other NLP techniques.
- Visualize and interpret clusters to uncover dominant themes in environmental reporting.
- Evaluate clustering performance using internal metrics such as Silhouette Score or Davies-Bouldin Index.

These insights gained from clustering can be used for thematic content analysis, trend monitoring and improved information accessibility.

1.3 Overview of the Dataset

The dataset, sourced from Kaggle, includes over 30,000 environmental news articles from *The Guardian*, covering topics like climate change, sustainability, pollution, and biodiversity. It contains five variables: title, introduction text, author, article text, and date published. This rich dataset supports tasks such as text clustering, sentiment analysis, and trend detection in environmental reporting.

2.0 Literature Review

2.1 Clustering Algorithms Review

Type of Algorithms	Description
Partition-Based Algorithms	Partitioning methods, such as Mini-Batch K-Means, Fuzzy C-Means, and standard K-Means, divide data into K clusters by optimizing similarity within clusters. These algorithms are fast, scalable, and suitable for large datasets but require predefined cluster numbers and are sensitive to initialization. They work best with spherical, equally sized clusters and struggle with complex data structures.
Hierarchical Clustering Algorithms	Hierarchical clustering, which includes agglomerative (bottom-up) and divisive (top-down) methods, doesn't require predefined clusters. It uses linkage methods (e.g., single, complete) to measure distances and visualizes results with dendrograms. While it offers a structured view and is suitable for small to medium datasets, it is computationally expensive and sensitive to linkage methods (Johnson, 1967).
Density-Based Algorithms	Density-based methods like DBSCAN and OPTICS identify clusters based on point density, allowing detection of arbitrary-shaped clusters and handling noise separately. Key parameters include ϵ (neighborhood radius)

	and minPts (minimum points for a cluster). These algorithms are effective for complex datasets but may struggle with varying densities (Ester et al., 1996).
--	--

K-Means Clustering

K-Means Clustering is a widely used, efficient algorithm that partitions data into K clusters by minimizing intra-cluster variance (MacQueen, 1967). It assigns points to the nearest centroid and updates centroids iteratively until convergence. While simple and scalable, it is sensitive to initial centroid placement and assumes spherical clusters (Jain, 2010).

Mini-Batch K-Means

Mini-Batch K-Means improves speed and memory usage by processing small data batches instead of the full dataset, making it ideal for large-scale and real-time applications (Sculley, 2010). However, it may yield less stable results due to the randomness of mini-batch selection (Bottou & Bousquet, 2008).

Fuzzy C-Means (FCM)

Fuzzy C-Means (FCM) is a clustering algorithm that allows data points to belong to multiple clusters using fuzzy logic (Bezdek, 1981). Instead of assigning each point to a single cluster, FCM computes membership degrees, showing how strongly each point is associated with every cluster. It updates cluster centers and memberships iteratively, guided by a fuzziness parameter (m), which controls the degree of cluster overlap (Dunn, 1973). FCM is especially useful for complex, overlapping data in domains like medical imaging, pattern recognition, and speech processing (Pham, Xu, & Prince, 2000), but it is computationally intensive and less suited for large datasets.

Spectral Clustering

Spectral Clustering is a graph-based clustering method that identifies complex cluster structures by transforming data into a graph and analyzing the eigenvectors of its Laplacian matrix (Ng, Jordan, & Weiss, 2001). Unlike traditional methods, it excels at detecting non-convex and arbitrarily shaped clusters. Spectral Clustering is widely used in image segmentation, social network analysis, and bioinformatics due to its ability to uncover meaningful patterns in high-dimensional or graph-structured data. However, its reliance on eigenvector computation makes it computationally expensive for large datasets, and its performance is sensitive to how the similarity graph is constructed.

Mean-Shift Algorithm

Mean-Shift Algorithm is a non-parametric, density-based clustering technique that identifies clusters by iteratively shifting data points toward regions of higher density. Unlike K-Means, which relies on distance metrics, Mean-Shift works by detecting modes (high-density areas) in the data distribution without requiring prior knowledge of the number of clusters. The algorithm uses Kernel Density Estimation (KDE) with a kernel function (usually Gaussian) to estimate the data density and then shifts points toward the mode of their local density function. The size of the neighborhood is controlled by the bandwidth parameter (h), which significantly impacts clustering results.

Sentence-BERT (SBERT)

SBERT is a deep learning-based method that generates sentence embeddings for clustering tasks. It builds on BERT and uses Siamese and triplet networks to optimize sentence similarity. SBERT is effective in semantic search and document clustering but is computationally intensive.

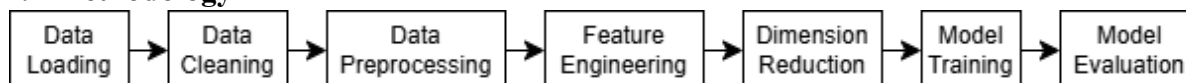
Latent Dirichlet Allocation (LDA)

LDA is a probabilistic topic modeling algorithm that identifies hidden topics in documents, assuming each document is a mixture of topics. It's widely used for topic modeling but requires hyperparameter tuning and struggles with short-text data.

Latent Semantic Analysis (LSA)

LSA uses Singular Value Decomposition (SVD) for dimensionality reduction and reveals hidden relationships between words and documents. It's applied in document similarity and search engines but has limitations in interpretability and polysemy handling.

2.2 Methodology



Data Loading: Import the dataset into a suitable data structure.

Data Cleaning: Clean the dataset such as removing missing values and duplicate values.

Data Preprocessing: Perform NLP techniques such as text cleaning, remove stopwords, tokenization, stemming and lemmatization.

Feature Engineering: Apply vectorization such as TF-IDF to convert the text data into numerical vectors that capture the importance of words in the document and word embedding.

Dimension Reduction: Reduces the number of features in the dataset while retaining key information.

Model Training: Select some suitable machine learning algorithms and use the preprocessed data to train the model.

Model Evaluation: Assesses the performance of the trained model using appropriate metrics.

3.0 Model Training

3.1 Data Cleaning

In this part, we perform the feature selection and data cleaning after data loading. We finally decided to choose the “title”, “intro text” and “article text” to perform clustering. The goal of the clustering is to identify the similar news articles based on its content.

	Title	Intro Text	Article Text
0	Liz Truss 'will approve more oil drilling if ...	Tory leadership candidate criticised by campai...	Liz Truss will sign off on a push for more oil...
1	Renewed Highland golf course plan has environm...	Scottish government rejected a new links at Co...	It is an area so tranquil that the notion of b...
2	Visiting green spaces deters mental health dr...	Positive effects were stronger among those rep...	Visits to parks, community gardens and other u...
3	Bought too much red cabbage? Turn it into a fe...	This fantastic vegan centrepiece makes full us...	I devised today's nut roast for Oddbox, a veg...
4	'This year has been very good': readers' UK bu...	Readers share their favourite sightings over t...	'Constant companions to our gardening/A peacoc...

→

	article
0	Liz Truss 'will approve more oil drilling if ...
1	Renewed Highland golf course plan has environm...
2	Visiting green spaces deters mental health dr...
3	Bought too much red cabbage? Turn it into a fe...
4	'This year has been very good': readers' UK bu...

After that, we remove the missing value and duplicated article.

<pre># Remove the missing value df = df.dropna() df.head()</pre>	<pre>df = df.drop_duplicates() df.head()</pre>																								
<table> <tr><th></th><th>article</th></tr> <tr><td>0</td><td>Liz Truss 'will approve more oil drilling if ...</td></tr> <tr><td>1</td><td>Renewed Highland golf course plan has environm...</td></tr> <tr><td>2</td><td>Visiting green spaces deters mental health dr...</td></tr> <tr><td>3</td><td>Bought too much red cabbage? Turn it into a fe...</td></tr> <tr><td>4</td><td>'This year has been very good': readers' UK bu...</td></tr> </table>		article	0	Liz Truss 'will approve more oil drilling if ...	1	Renewed Highland golf course plan has environm...	2	Visiting green spaces deters mental health dr...	3	Bought too much red cabbage? Turn it into a fe...	4	'This year has been very good': readers' UK bu...	<table> <tr><th></th><th>article</th></tr> <tr><td>0</td><td>Liz Truss 'will approve more oil drilling if ...</td></tr> <tr><td>1</td><td>Renewed Highland golf course plan has environm...</td></tr> <tr><td>2</td><td>Visiting green spaces deters mental health dr...</td></tr> <tr><td>3</td><td>Bought too much red cabbage? Turn it into a fe...</td></tr> <tr><td>4</td><td>'This year has been very good': readers' UK bu...</td></tr> </table>		article	0	Liz Truss 'will approve more oil drilling if ...	1	Renewed Highland golf course plan has environm...	2	Visiting green spaces deters mental health dr...	3	Bought too much red cabbage? Turn it into a fe...	4	'This year has been very good': readers' UK bu...
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Since there are various types and forms of the article, we considered those articles with extremely short or long paragraphs and higher or lower repetition words as outliers.

```
# Remove extremely short or long documents
df["Text_Length"] = df["article"].apply(lambda x: len(x.split()))

# Define length thresholds
min_length = df["Text_Length"].quantile(0.2)
max_length = df["Text_Length"].quantile(0.8)

df = df[(df["Text_Length"] > min_length) & (df["Text_Length"] < max_length)]
df.head()
```

	article	Text_Length
2	Visiting green spaces deters mental health dr...	593
3	Bought too much red cabbage? Turn it into a fe...	542
4	'This year has been very good': readers' UK bu...	941
6	Scaly stowaway; lizard makes an appearance at...	657
8	UK energy industry urges ministers to stick w...	806

```
# Remove documents with high word repetition
# Detect spam-like text
def word_repetition(text):
    words = text.split()
    word_counts = Counter(words)
    most_common_word, most_common_count = word_counts.most_common(1)[0]
    return most_common_count / len(words)

df["Repetition_Ratio"] = df["article"].apply(word_repetition)
df = df[(df["Repetition_Ratio"] < 0.06) & (df["Repetition_Ratio"] > 0.045)]
df.head()
```

	article	Text_Length	Repetition_Ratio
3	Bought too much red cabbage? Turn it into a fe...	542	0.047970
6	Scaly stowaway; lizard makes an appearance at...	657	0.059361
8	UK energy industry urges ministers to stick w...	806	0.050868
16	This article is more than 11 months oldFood, ...	765	0.045752
17	This article is more than 4 months oldUS behi...	902	0.057650

3.2 Data Preprocessing

In this stage, we will perform NLP techniques which are text cleaning, tokenization, stemming and lemmatization.

Stemming Results >

```
3      [bought, much, red, cabbag, turn, festiv, nut,...
6      [scali, stowaway, lizard, make, appear, chelse...
8      [uk, energi, industri, urg, minist, stick, net...
16     [articl, month, oldfood, feed, fuel, global, s...
17     [articl, month, oldu, behind, third, global, o...
...
30038  [capit, part, solut, climat, crisi, say, mark,...
30040  [cours, hot, return, wrestl, eastend, legendal...
30042  [countri, diari, seal, remain, show, natur, se...
30050  [bhp, face, fresh, call, quit, miner, council,...
30054  [climat, chang, could, make, insur, expens, pe...
Name: stem_article, Length: 8187, dtype: object
```

Lemmatizing Results >

```
3      buy much red cabbage turn festive nut roast re...
6      scaly stowaway lizard make appearance chelsea ...
8      uk energy industry urge minister stick net zer...
16     article months oldfood fee fuel global seaweed...
17     article months oldus behind third global oil g...
...
30038  capitalism part solution climate crisis say ma...
30040  course hot return wrestle eastenders legendale...
30042  country diary seal remain show nature sentimen...
30050  bhp face fresh call quit minerals council ahea...
30054  climate change could make insurance expensive ...
Name: article, Length: 8187, dtype: object
```

3.3 Model Training Environment and Tools

In this part, we develop our project using Python due to its rich ecosystem of machine learning libraries and community support. We employ various types of libraries to implement the algorithms.

Tools	Explanation
Numpy	Perform numeric operations on arrays and metrics
Pandas	Data preprocessing and manipulation
Matplotlib & Seaborn	Visualization of clustering results and model performance
nltk	Text processing, tokenization, stemming and other NLP task
Scikit-learn	Used to implement several unsupervised learning algorithms 1. Spectral Clustering 2. Mean Shift 3. MiniBatch K-Means
Ast	Convert strings to Python objects
t-SNE	Reduces high-dimensional data to 2D or 3D for easy viewing
TruncatedSVD	Linear dimensionality reduction method used for sparse text data
UMAP	Advanced dimensionality reduction and visualization tool
Spectral Clustering	Finds groups in data using graph-based methods
KMeans	Splits data into groups (clusters) based on similarity
torch	Build and train deep learning models
LatentDirichletAllocation	Finds hidden topics inside a group of documents.
SentenceTransformer	Compute the sentence and paragraph embeddings
fuzzy-c-means	Used to implement the fuzzy c means model

Development Environment	Platform
IDE / Editor	Visual Studio Code / Jupyter Notebook
Environment Management	Anaconda for package and environment isolation

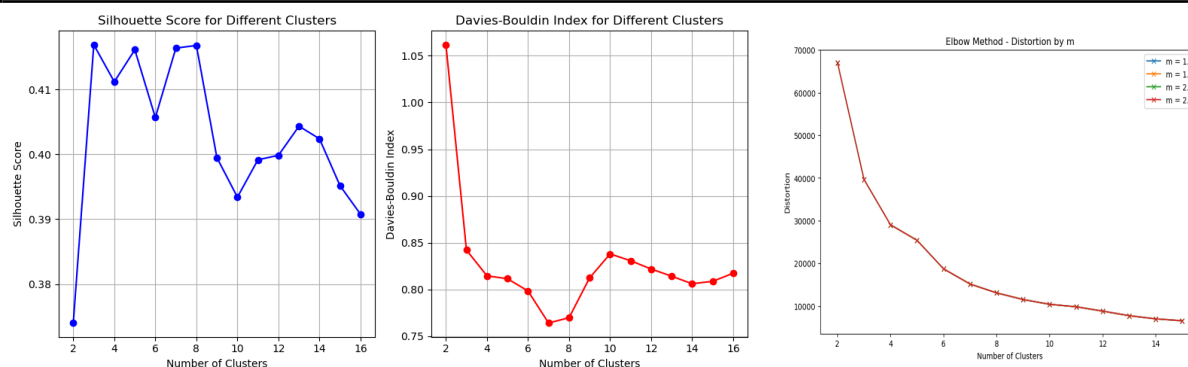
3.4 Model Training

3.4.1 Fuzzy C Means

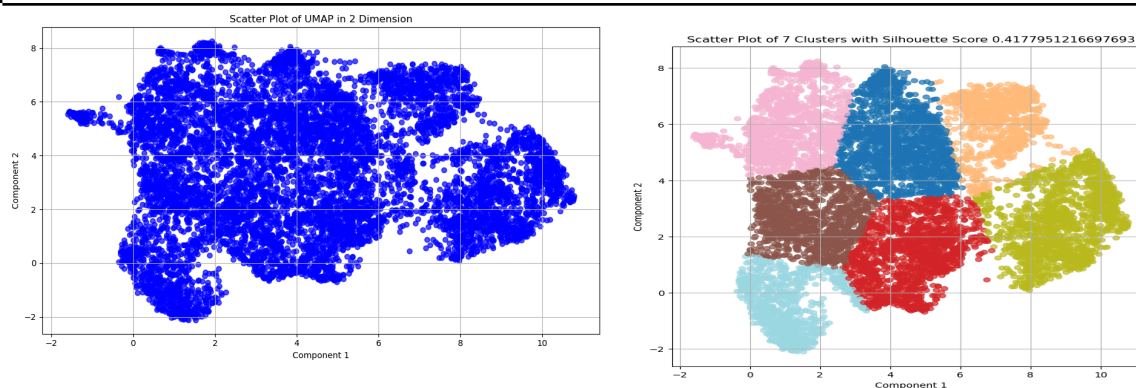
Fuzzy C-Means (FCM) algorithm was trained using dense semantic embeddings derived from a pretrained BERT (bert-base-uncased) model. Each article was transformed into a contextualized 768-dimensional embedding using the CLS token output. These embeddings were then standardized using StandardScaler. Due to the high dimensionality, a two step dimensionality reduction approach was applied.

TruncatedSVD reduced the data to 50 dimensions while preserving key variance components.

UMAP was used to further reduce the dimensionality to 2 for visualization, using cosine distance as the metric. This approach preserves both global and local structures in the data.



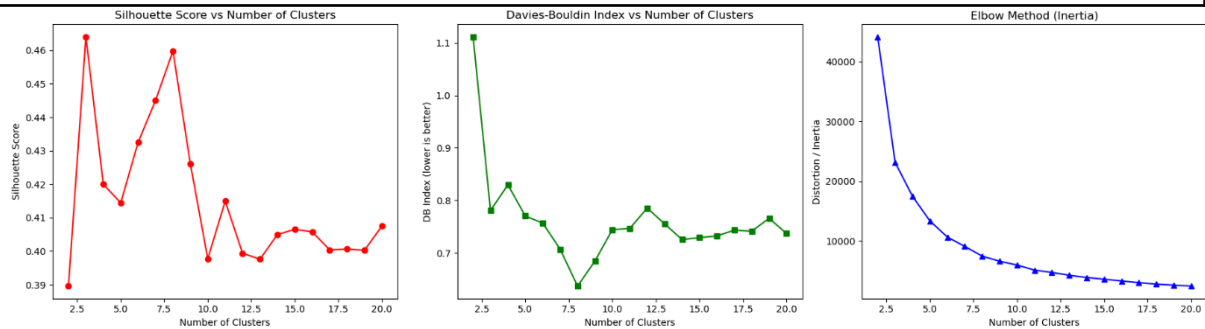
The diagram presents evaluation metrics for different numbers of clusters using the Silhouette Score, Davies-Bouldin Index, and Elbow Method. The Silhouette Score, which measures how similar an object is to its own cluster compared to others (higher is better), suggests optimal clustering between 3 and 8 clusters, excluding 6. The Davies-Bouldin Index, where lower values indicate better-defined clusters, points to 7 and 8 clusters as the most suitable. The Elbow Method, which looks for a point where adding more clusters yields diminishing returns, shows a clear drop at 5 clusters. These results were compared to determine the most suitable number of clusters for segmentation.



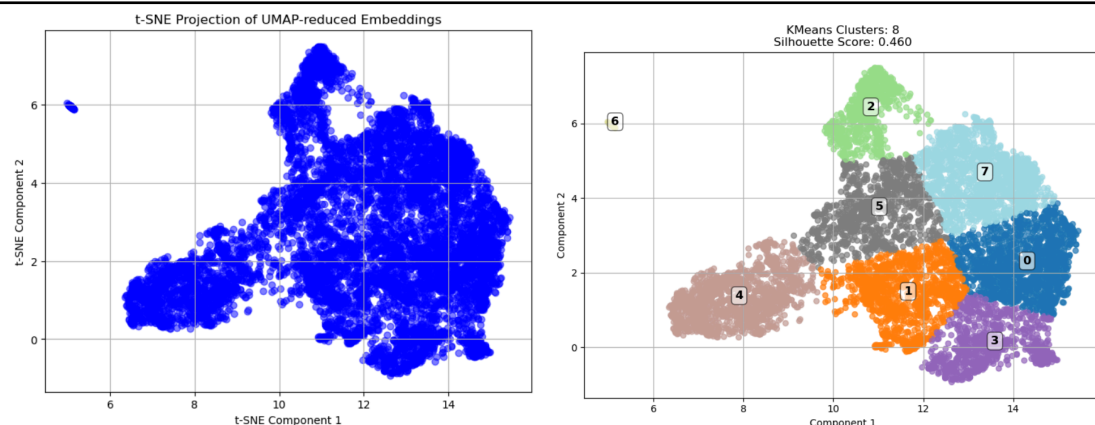
The figure above presents two visualizations of the dataset after applying UMAP for dimensionality reduction. The left plot shows a 2D scatter plot of the data points without any clustering labels. UMAP was used to project the high-dimensional feature space into two dimensions, preserving the overall structure and similarity relationships within the data. Each blue point represents a sample, and the distribution suggests the presence of several distinct groupings, although no cluster assignment has been made at this stage. The right plot shows the result of applying fuzzy c-means clustering to the same UMAP-reduced data. The data was clustered into seven groups, with each color representing a different cluster. In fuzzy c-means, each data point has a degree of belonging to multiple clusters. However, for visualization purposes, each point is assigned to the cluster for which it has the highest membership score. The clustering performance is evaluated using the Silhouette Score, which in the case is 0.418. The score indicates a moderate clustering quality, suggesting that the data exhibits some meaningful cluster structure, although they may be some overlaps between clusters.

3.4.2 Latent Semantic Analysis with K-Means

Latent Semantic Analysis (LSA) is a dimensionality reduction technique based on matrix factorization, typically implemented using the TruncatedSVD class from Scikit-learn's decomposition module. In text clustering, it is used to project high-dimensional embeddings, such as those from BERT into a lower-dimensional semantic space. This process captures the latent structure of the text data, highlighting key semantic relationships while reducing noise. It is first applied to reduce dimensionality, followed by KMeans clustering to group the documents. By combining LSA with contextual embeddings, this approach enables more efficient and semantically meaningful clustering, especially when dealing with large, complex text datasets.



The diagram shows clustering evaluation using three metrics, which are the Silhouette Score, Davies-Bouldin Index (DBI), and the Elbow Method (Inertia) across a range of 2 to 20 clusters to determine the optimal number of clusters for article text clustering. The Silhouette Score, which measures how well data points fit within their assigned clusters, peaks at 3 clusters with a score of 0.464, followed closely by 8 clusters at 0.460, suggesting both configurations offer strong separation and internal cohesion. After 8 clusters, the silhouette score begins to slightly decline and fluctuate, indicating diminishing clarity in cluster separation. The Davies-Bouldin Index, where lower values indicate better clustering, reaches its lowest point at 8 clusters with a score of 0.637, supporting this configuration as optimal in terms of compactness and separation. The Elbow Method, visualized using Inertia, shows a steep decline from 2 to 8 clusters, after which the rate of decrease slows noticeably. This indicates that the most substantial improvement in within-cluster compactness occurs within this range, with 8 clusters presenting a strong “elbow point.” Beyond this, the reduction in inertia becomes marginal, implying limited gain from adding more clusters. Taken together, these metrics suggest that while 3 clusters offer slightly higher silhouette performance, 8 clusters strike the best balance across all three metrics. This configuration achieves solid separation, compactness, and efficient cluster definition, making it a strong candidate for the final clustering model.

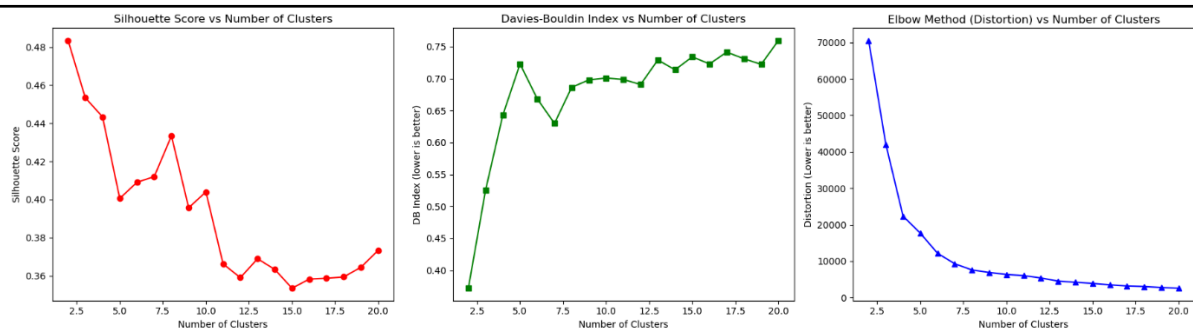


The left diagram displays the data distributed after reduced embeddings. Principal Component Analysis (PCA) using TruncatedSVD is implemented to reduce the embeddings to 50 components, keeping most of the data's variance. Then, UMAP further reduces it to 2 dimensions using 30 neighbors and cosine distance, allowing for easier clustering and visualization while preserving the data's local structure. The second diagram is about the KMeans clustering visualization performed

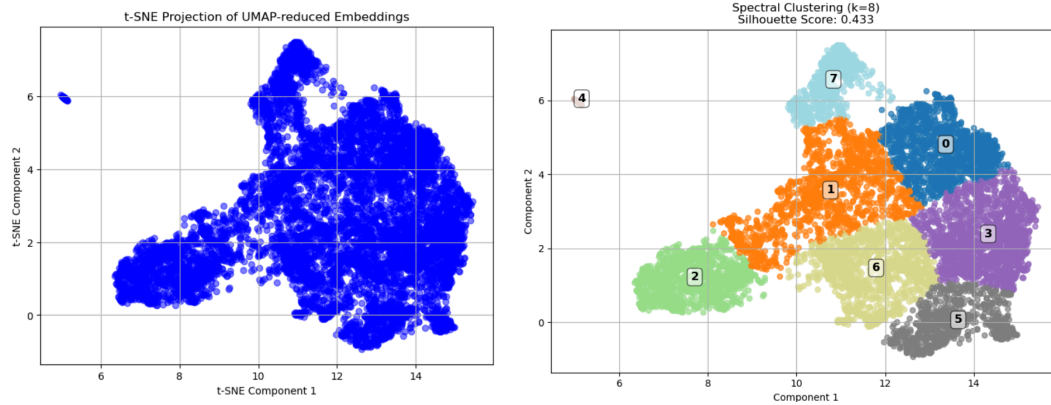
with $k=8$, based on TruncatedSVD and UMAP dimensionality reduction, reveals distinct groupings with generally well-separated clusters. Clusters 0, 1, 3, 4, 5, and 7 form the bulk of the dataset, each containing between 900 and 1,446 data points, indicating strong representation of major themes or patterns. Cluster 2, with 700 points, is compact and distinct in the upper region of the plot, suggesting a coherent subgroup. Notably, Cluster 6, with only 58 data points, is clearly an outlier both visually and numerically, positioned far from the dense cluster formations and representing a potentially unique or anomalous subset of the data. Cluster 5 which are located centrally, overlaps with several surrounding clusters and may represent transitional or mixed-type entries. The clustering quality is supported by performance metrics, which are a Silhouette Score of 0.460 indicates moderate cohesion and separation, a Davies-Bouldin Index of 0.637 reflects good cluster compactness and separation, and a Calinski-Harabasz Index of 10279.292 highlights strong intra-cluster density and inter-cluster separation. These results confirm that the clustering successfully captures both dominant structures and rare edge cases in the dataset.

3.4.3 Spectral Clustering

Spectral clustering is a graph-based algorithm, implemented via the SpectralClustering class from Scikit-learn's cluster module. In text clustering, it is often paired with word embeddings like those from BERT to better capture semantic meaning. The BERT tokenizer, trained on English text, converts raw input into tokenized numerical form, while the pre-trained BERT model processes this input to generate contextual embeddings. Unlike TF-IDF or bag-of-words, these embeddings offer rich semantic representations, enabling spectral clustering to group documents based on deeper linguistic similarity.



The diagram shows clustering evaluation using three metrics: Silhouette Score, Davies-Bouldin Index (DBI), and the Elbow Method (Distortion), across 2 to 20 clusters. These metrics help determine the optimal number of clusters for article text clustering. The Silhouette Score measures how well each point fits within its cluster, with higher scores indicating better-defined clusters. The highest silhouette score which is 0.483 occurs at 2 clusters, suggesting the most distinct separation is achieved there. As the number of clusters increases, the score declines, indicating less clearly separated groups. The Davies-Bouldin Index, where lower values are better, also reaches its best value (0.372) at 2 clusters. This implies strong compactness and separation at that point. The DBI values increase and fluctuate, showing reduced clustering quality with more clusters. The Elbow Method examines within-cluster distortion and identifies an "elbow" point around 4 to 6 clusters, where adding more clusters yields diminishing returns. Taken together, these results suggest that while 2 clusters offer the best separation, 4 to 6 clusters may provide a practical balance between detail and efficiency.

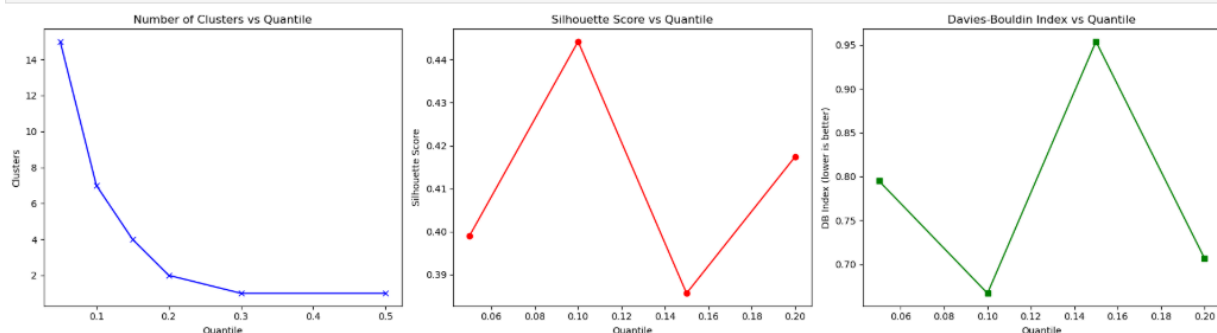


The left diagram displays the data distributed after reduced embeddings. The Principal Component Analysis (PCA), using TruncatedSVD, reduces the dimensionality of the embeddings to 50 components, simplifying the data while retaining most of its variance. This transformation projects the data into a lower-dimensional space. Next, Uniform Manifold Approximation and Projection (UMAP) further reduces the dimensionality to 2 components for clustering and visualization. UMAP is configured with 30 neighbors and the cosine distance metric to maintain the local structure of the data. This 2-Dimensional representation facilitates easier visualization and analysis of the clusters.

Cluster number 8 is chosen for Spectral Clustering, resulting in a well-balanced segmentation of the dataset. The Silhouette Score of 0.433 indicates reasonable separation and cohesion within clusters, while the Davies-Bouldin Index of 0.686 suggests good intra-cluster compactness and inter-cluster separation. The high Calinski-Harabasz Index of 9075.092 further confirms well-defined clusters. The diagram reveals that Clusters 0, 1, 3, and 6 represent the core of the dataset, while Clusters 2, 5, and 7 are smaller but still distinct. Cluster 4, which consists of only 58 points, appears as an isolated group, likely indicating outliers or a specialized subset. Cluster 2 is compact and well-separated, suggesting it forms a coherent subgroup with unique characteristics. Meanwhile, Cluster 7 shows noticeable separation, indicating it represents another distinct population within the data.

3.4.4 Mean Shift

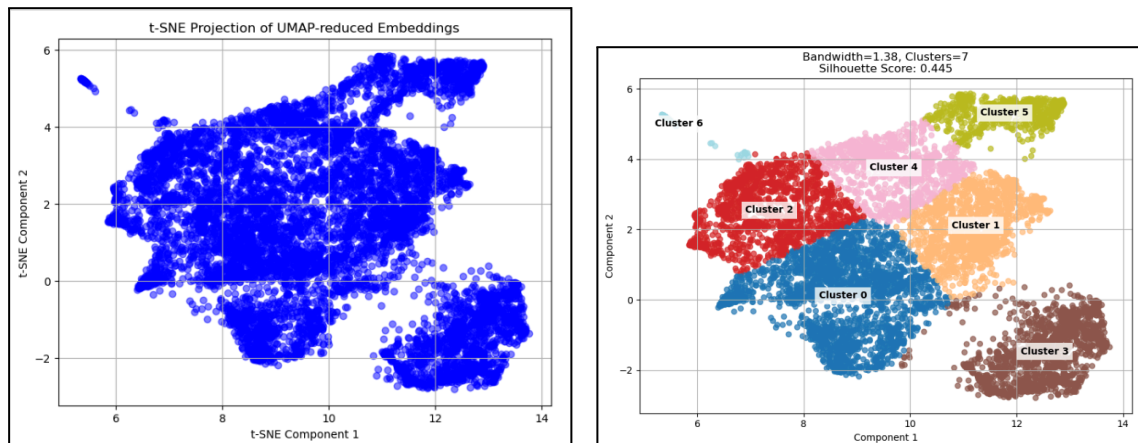
The Mean Shift clustering algorithm, a non parametric density based method was employed to identify natural groupings within a dataset by leveraging the Mean shift class from the scikit-learn library. This approach was applied to UMAP reduced embeddings, aiming to uncover meaningful clusters while evaluating their quality through various metrics and visualizations.



The three line graphs indicate the clustering evaluation metrics, used to determine the optimal number of clusters for a dataset by varying a parameter called Quantile. The first diagram is “Number of Clusters vs Quantile”. As the quantile increases from 0.06 to 0.20, the number of clusters decreases sharply from around 14 to 1. This indicates that higher quantile values result in fewer clusters, possibly because the clustering algorithm becomes more selective on grouping data points. The middle diagram is “Silhouette Score vs Quantile”. The silhouette score measures how well separated clusters are. The higher the score is better with a maximum of 1. From the image, the score starts at 0.40 at quantile 0.06, peaks at 0.44 around quantile 0.10, dips to 0.38 at quantile 0.14 and rises to 0.42 at quantile 0.18. The peak at 0.10 suggests that this quantile value might yield the best clustering quality in terms of separation and cohesion. The last diagram is “Davies-Bouldin Index vs Quantile”. The DB Index measures the average similarity ratio of each cluster with its most

similar cluster. The lower is better with 0 being ideal. Based on the diagram, the DB Index starts at 0.80 at quantile 0.06, drops to 0.65 around quantile 0.14 and rises again to 0.95 at quantile 0.20. The lowest value 0.65 at quantile 0.14 indicates that this quantile might produce the best clustering in terms of minimizing inter cluster similarity.

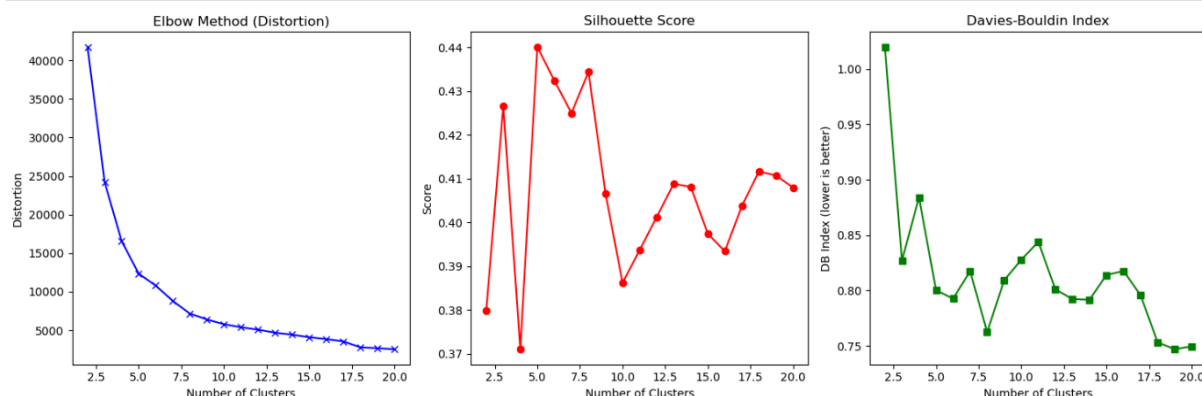
For the Mean Shift clustering algorithm, the optimal quantile lies between 0.10 and 0.14. A quantile of 0.10 maximizes the Silhouette Score (0.44), indicating better cluster separation and cohesion, while a quantile of 0.14 minimizes the Davies-Bouldin index (0.65), suggesting less inter cluster similarity. However, the number of clusters drops significantly from 14 to 1 as the quantile increases from 0.06 to 0.20, so a quantile around 0.10 might be preferred to maintain a reasonable number of clusters while optimizing quality.



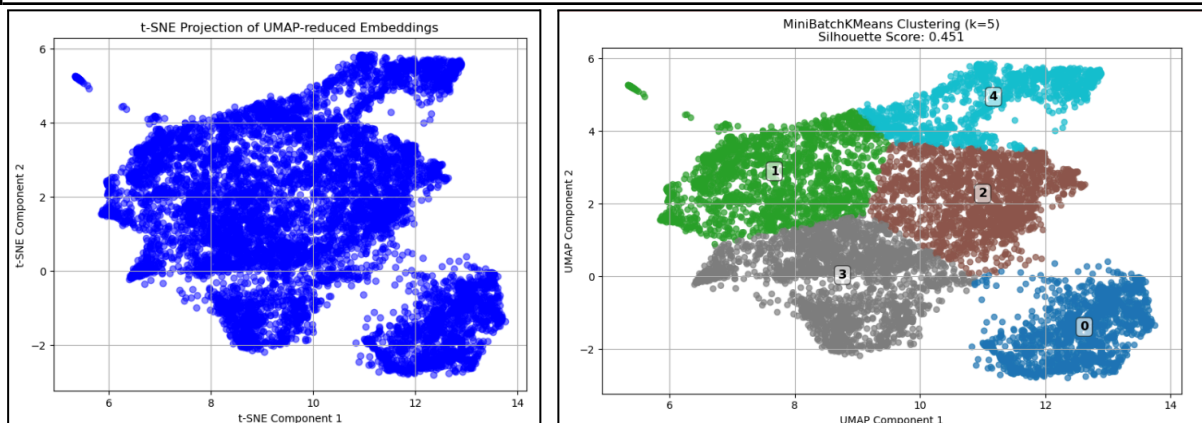
The first diagram shows a t-SNE projection of the UMAP reduces embedding. All points are in blue indicating no clustering applied here. The t-SNE projection shows a similar structure to the UMAP scatter plot, with dense regions corresponding to the cluster identified by Mean Shift. t-SNE tends to emphasize local structure more than UMAP, often creating tighter clusters in the visualization. However, the overall layout is consistent with the UMAP plot, confirming that the dense regions identified by Mean Shift are natural modes in the data. The central region appears slightly more scattered in t-SNE, reinforcing the idea that this area has a more continuous density, making it harder for Means Shift to define a distinct cluster. Next, the second diagram is a scatter plot visualizing UMAP reduced data points in 2D space with points colored according to the clusters assigned by Mean Shift with bandwidth 1.38, resulting in 7 clusters with a Silhouette Score of 0.445. Most clusters are relatively well separated with clear boundaries, except for Cluster 4 in the central region which is more scattered, aligning with the moderate separation indicated by the Silhouette Score. The bandwidth of 1.38 controls the kernel radius for density estimation in Mean Shift. A smaller bandwidth would yield more clusters, while a larger one would merge them, as seen in the earlier graphs where higher quantiles reduced the number of clusters. With 7 clusters, this bandwidth achieves balanced results.

3.4.5 K-Means Mini Batch

Mini Batch K-Means is a scalable variant of the traditional K-Means clustering algorithm, designed to efficiently handle large datasets. It works by randomly sampling small batches of data to update cluster centroids iteratively rather than using the entire dataset in each iteration. Mini Batch K-Means is particularly useful for applications like text clustering, where datasets can be large and it supports online learning by adapting to new data over time.



These diagrams show the three plots evaluating the optimal number of clusters for MiniBatch K-Means using different metrics which are the Elbow Method (Distortion), Silhouette Score and Davies-Bouldin Index. Each plot accesses quality across a range of cluster numbers (2 to 20). The left plot which is Elbow Method (Distortion) displays distortion as the number cluster increases. It starts high and decreases sharply until around 5 clusters, then levels off indicating diminishing returns in reducing distortion with more clusters. The “elbow” around 5 clusters suggests this as a potential optimal number, balancing cluster compactness and model complexity. The middle plot shows the Silhouette Score, which measures how similar points are within the cluster compared to other clusters. (The score higher is better, ranging from -1 to 1). The score fluctuates between 0.37 and 0.44, peaking around 4 clusters (0.43) and showing another high point around 8 clusters. This suggests that 4 or 8 clusters might provide better separation and cohesion than the 5 clusters used in your original model. The right plot presents the Davies-Bouldin Index, which evaluates cluster separation and compactness; The index starts high, drops sharply to 0.80 around 5 clusters and fluctuates thereafter with lower values around 10 and 17 clusters. A lower DB Index at 5 clusters supports the elbow method’s indication, but the variability suggests other cluster counts might also be viable. The Elbow Method and Davies-Bouldin Index both point to around 5 clusters as a reasonable choice, aligning with your original model. However, the Silhouette Score suggests 4 or 8 clusters might offer better cluster quality.



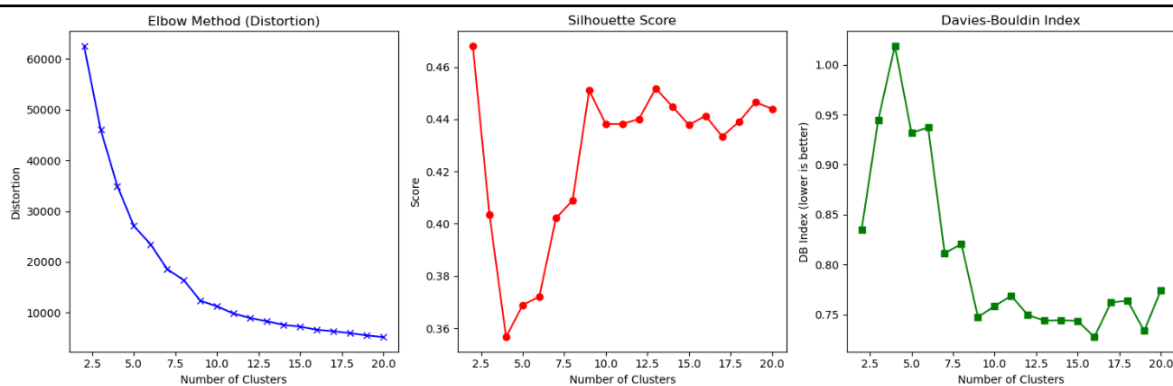
The first image presents a scatter plot that visualizes the 2D projection of UMAP-reduced embeddings using t-SNE, a technique that emphasizes local structure to reveal patterns in high-dimensional data. Each blue dot represents one of the 8000 new articles, distributed in a 2D space after applying t-SNE on top of the UMAP reduced embedding. The plot reveals several dense clusters of points indicating potential natural groupings based on their semantic content of the articles, alongside sparse regions and a few outliers, such as the lone point at the top left. While the dense clusters suggest that the data is suitable for clustering, the lack of color based or labeled clusters limit the deeper analysis, providing only a raw view of the data’s structure post t-SNE. In contrast, the second image visualizes the clustering results of MiniBatch K-Means applied on the same UMAP reduced embeddings, grouping the 8000 articles into 5 clusters. The points are color coded by cluster (0 to 4) with cluster centers labeled to show the mean position of each cluster. The silhouette score of 0.451 indicates moderate cluster quality, as values closer to 1 indicate better-defined clusters. The clusters are relatively well separated, though some overlap exists and

the distribution of points highlights varying thematic densities. These visualizations confirm that the news articles exhibit meaningful semantic groupings, and evidenced by the dense clusters in the t-SNE projection and the structured clustering in the MiniBatchKMeans results.

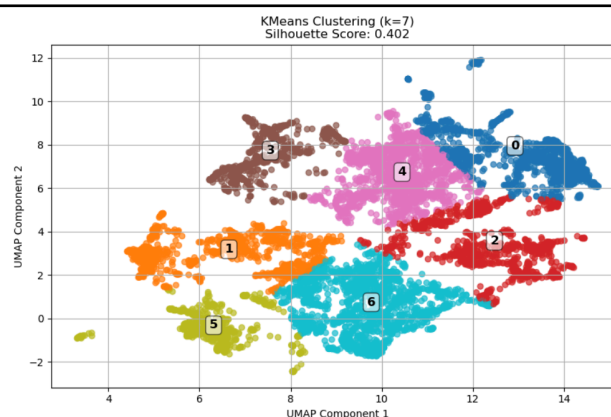
3.4.6 K-Means with LDA & Sentence-BERT hybrid

Sentence-BERT is used to generate high-quality dense vector embeddings for entire sentences or documents by leveraging transformer-based models. These embeddings effectively capture the semantic meaning of the text. In contrast, Latent Dirichlet Allocation (LDA) is a topic modeling algorithm that identifies latent topic distributions within a corpus, enabling the extraction of dominant themes from documents.

By combining Sentence-BERT and LDA, it is possible to enrich the feature space with both semantic and topical information. This fusion enhances the performance of clustering algorithms, such as K-Means, by allowing them to group documents that are not only semantically similar but also share common topics.



The diagram above represents the Elbow Method (Distortion), Silhouette Score, and Davies-Bouldin Index, which are commonly used to evaluate the optimal number of clusters in a dataset. The Elbow Method helps determine the ideal number of clusters by identifying the point where adding more clusters no longer significantly reduces the distortion. In the graph, the distortion value drops rapidly at first and then begins to level off. The curve starts to flatten around $k = 6$ to 8 , indicating the "elbow point." Therefore, choosing between 6 and 8 clusters is considered a good choice based on this method. The Silhouette Score is used to measure the cohesion and separation of clusters. From the graph, the score increases sharply and peaks around $k = 7$, after which it fluctuates. The highest Silhouette Score appears at $k = 7$, indicating that 7 clusters provide the most compact and well-separated grouping. The Davies-Bouldin Index evaluates how compact and distinct the clusters are. In the graph, the index drops sharply around $k = 7$ and then remains relatively stable. This trend confirms that using 7 clusters offers a good balance between tight intra-cluster similarity and clear inter-cluster separation.



The cluster diagram presents the results of KMeans clustering ($k=7$) applied to a hybrid feature space combining Sentence-BERT embeddings and LDA topic distributions, projected into two dimensions using UMAP for visualisation. A text sample is represented by each point in the plot,

which is arranged according to how similar it is to other points in terms of subject and semantics. The data was grouped into seven clusters (labelled 0 to 6), each indicated by a distinct colour and marked with its cluster number at the centroid. Cluster 0 is distinct and moderately spread out. Cluster 1 is compact and exhibits strong internal coherence. Cluster 2 is elongated and shows slight overlaps with neighboring clusters. Cluster 3 is compact and well-separated from the others. Cluster 4 is centrally located and overlaps widely with multiple clusters. Cluster 5 is tight and represents a niche grouping. Cluster 6 is broad and diverse in its composition. The silhouette score of 0.402 reflects moderately effective clustering, with clear structure in some areas but room for improvement in others. The overlap between certain clusters suggests either topic blending or fuzzy semantic boundaries. This setup effectively balances semantic richness (via SBERT) with topic clarity (via LDA), providing a strong foundation for exploring patterns in the text.

4.0 Performance Evaluation

4.1 Evaluation Metrics

4.1.1 Silhouette Score

The silhouette score evaluates how well each data point fits within its assigned cluster, compared to other clusters. It ranges from -1 to 1, where a higher value indicates that the object is well-matched to its own cluster. It balances both cohesion and separation.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$a(i)$ = average distance between i and all other points in the same cluster

$b(i)$ = smallest average distance between i and all point in any other cluster

4.1.2 Davie-Bouldin Index

The Davie-Bouldin Index measures the clustering quality based on the ratio within cluster distance to between cluster distance. Lower values indicate a better clustering, as it means the clusters are compact and well-separated. It is useful for comparing different clustering solutions, especially when the number of clusters varies.

$$DB = \frac{1}{k} \sum_{i=1}^k \max \left(\frac{\Delta(X_i) + \Delta(X_j)}{\delta(X_i, X_j)} \right)$$

ΔX_k = the intracluster distance within the cluster X_k

$\delta(X_i, X_j)$ = the intercluster distance between the clusters X_i and X_j .

4.1.3 Calinski-Harabasz Index

The Calinski-Harabasz Index calculates the performance based on the ratio of the spread between cluster dispersion to within cluster dispersion. A higher value indicates well-separated and dense clusters, suggesting a better defined grouping. It is especially effective when comparing different clustering configurations.

$$CH = \frac{BCSS/(k - 1)}{WCSS/(n - k)}$$

$BCSS$ (Between-Cluster Sum of Squares) = the weighted sum of the squared Euclidean distances from each cluster's centroid to the overall mean of the entire dataset.

$WCSS$ (Within-Cluster Sum of Squares) = the total of the squared Euclidean distances between each data point and the centroid of the cluster it belongs to.

4.2 Comparison of Model Performance

	Number of Cluster	Silhouette Score	Davie-Boudin Index	Calinski-Harabasz Index
Fuzzy C Means	7	0.4178	0.7613	-
Spectral Clustering	8	0.433	0.686	9075.092
Latent Semantic Analysis with KMeans	8	0.460	0.637	10279.292
Mean Shift	7	0.445	-	-
K Means Mini Batch	5	0.451	-	-
K-Means with LDA & Sentence-BERT hybrid	7	0.402	-	-

The algorithms highlighted in green are those shortlisted for the final discussion. They were selected based on their reasonable and optimal silhouette scores, as well as the compactness of their clusters in the scatter plot. These characteristics indicate good separation and consistency within clusters, making them suitable candidates for deeper analysis.

5.0 Pattern and Feature Analysis

5.1 Important Variables and Features

At the initial stage, the main focus was on the article column, which contained raw text data. No filtering of common or rare words had been performed yet. As a result, the text features included a mix of extremely frequent terms (e.g., stopwords) and rare or domain-specific words. This unfiltered state often caused high dimensionality and noise in the vectorized features, potentially diluting meaningful patterns.

5.2 Underlying Patterns and Decision Boundaries

Preliminary clustering and visualization of the data showed unclear boundaries between groups. The presence of overly common words made clusters form around irrelevant similarities, such as articles sharing generic phrases. Conversely, rare words introduced sparsity, making some samples outliers. These early observations indicated that the raw data needed further cleaning before meaningful clusters or decision boundaries could be identified.

6.0 Error and Bias Analysis

6.1 Error Examination

Errors at this point were mostly related to noise in the data. Articles with frequent stopwords or filler content skewed distance calculations, while articles with rare or unique vocabulary appeared isolated or incorrectly grouped. These errors affected the interpretability of cluster assignments and highlighted the need for frequency-based filtering.

6.2 Generalization and Bias

The dataset, in its raw form, showed bias towards high-frequency words that are commonly used across all documents. These words overwhelmed more informative terms, causing the clustering model to generalize poorly. Without addressing this imbalance, there was a risk of forming artificial clusters driven by word count rather than meaningful content.

7.0 Model Shortlisting

7.1 Final Justification and Discussion

UMAP was selected for dimensionality reduction due to its ability to preserve both local and global data structures in lower dimensions, outperforming methods like PCA and t-SNE in balancing speed and topology preservation. Reducing dimensionality before clustering helps mitigate the curse of dimensionality and improves cluster separability.

For clustering, Fuzzy C-Means was chosen over hard clustering algorithms such as K-Means, as it allows soft membership assignment. This is crucial for datasets where class boundaries are not well-defined, providing more nuanced insights. Compared to density-based methods like DBSCAN, Fuzzy C-Means is better suited for identifying overlapping clusters within continuous data distributions.

The UMAP + Fuzzy C-Means combination provided a meaningful cluster structure with a Silhouette Score of 0.418, demonstrating moderate but acceptable clustering quality for the purposes of this study.

PART B - Refining the Model

1.0 Hyperparameter Tuning

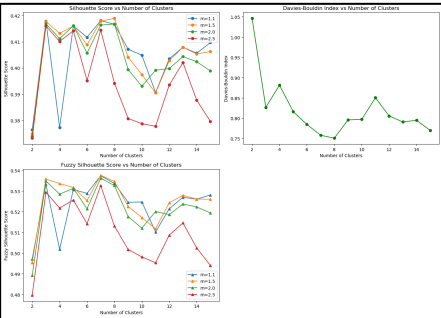
1.1 Tuning Techniques Used

To optimize model performance, hyperparameter tuning was carried out to identify the best combination of cluster count and fuzziness level. The key parameters tested were:

Number of Clusters (c)	2 to 15
Fuzziness Coefficient (m)	1.1, 1.5, 2.0, 2.5

A **grid search technique** was used to evaluate all (m, c) combinations. The FCM algorithm was run with a **maximum of 1000 iterations** and a **convergence threshold of 0.001**.

1.2 Visualization During Hyperparameter Tuning

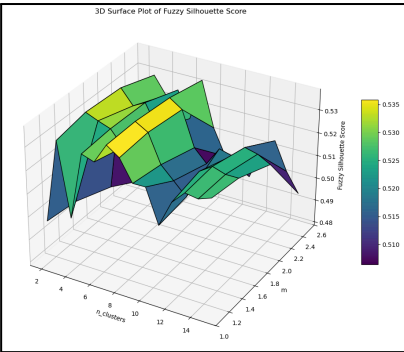


The figure consists of three line graphs arranged in a 2x2 grid, with the bottom-right cell empty. All graphs share an x-axis labeled 'Number of Clusters' ranging from 2 to 16.

- Silhouette Score vs Number of Clusters:** The y-axis ranges from 0.40 to 0.45. Four lines represent different values of m : $m=1.1$ (blue), $m=1.5$ (orange), $m=2.0$ (green), and $m=2.5$ (red). All lines show a general downward trend as the number of clusters increases, with a notable dip at 4 clusters.
- Davies-Bouldin Index vs Number of Clusters:** The y-axis ranges from 0.75 to 1.00. The same four lines for m are shown. The index generally decreases as the number of clusters increases, starting near 1.00 at 2 clusters and dropping to around 0.75-0.80 by 16 clusters.
- Fuzzy Silhouette Score vs Number of Clusters:** The y-axis ranges from 0.40 to 0.54. The same four lines for m are shown. The scores fluctuate between 0.45 and 0.54, with a general peak around 6-8 clusters.

From the graph, it shows the silhouette score, davies-bouldin index and also fuzzy silhouette scores with different numbers of clusters. Table below shows the optimal number of clusters:

m	Silhouette Score	Fuzzy Silhouette Score
1.1	5 to 8	5 to 8
1.5	3 to 8	3 to 8
2.0	3 to 8	3 to 8
2.5	3 to 7	3 to 8



A 3D surface plot showing the Fuzzy Silhouette Score as a function of the number of clusters ($n_clusters$) and the fuzziness coefficient (m). The x-axis represents $n_clusters$ from 2 to 16, and the y-axis represents m from 1.0 to 2.4. The z-axis represents the Fuzzy Silhouette Score, with a color bar on the right ranging from 0.48 to 0.535. The surface shows a prominent peak in the score when $n_clusters$ is between 6 and 8 and m is between 1.2 and 1.8.

From the graph above, we can see that the $n_cluster$ between 6 and 8, m between 1.2 to 1.8, the fuzzy silhouette score is quite optimal.

	m	n_clusters	silhouette_score	partition_coefficient	partition_entropy	fuzzy_silhouette_score
m	1.000000	0.000000	-0.231474	-0.843201	0.806986	-0.282830
n_clusters	0.000000	1.000000	-0.149166	-0.227206	0.068463	-0.123079
silhouette_score	-0.231474	-0.149166	1.000000	0.273999	-0.268618	0.876873
partition_coefficient	-0.843201	-0.227206	0.273999	1.000000	-0.953866	0.310446
partition_entropy	0.806986	0.068463	-0.268618	-0.953866	1.000000	-0.299838
fuzzy_silhouette_score	-0.282830	-0.123079	0.876873	0.310446	-0.299838	1.000000

From the table above, it shows the correlation of each of the parameters such as m and $n_cluster$ with the scores. The score between $silhouette_score$ and $fuzzy_silhouette_score$ is quite high. Thus, we can examine the best model based on the silhouette score and fuzzy silhouette score.

Top Silhouette Scores: <table><tr><th></th><th>m</th><th>n_clusters</th><th>silhouette_score</th></tr><tr><td>20</td><td>1.5</td><td>8.0</td><td>0.4189</td></tr><tr><td>5</td><td>1.1</td><td>7.0</td><td>0.4180</td></tr><tr><td>1</td><td>1.1</td><td>3.0</td><td>0.4178</td></tr></table> Top Partition Coefficient (PC): <table><tr><th></th><th>m</th><th>n_clusters</th><th>partition_coefficient</th></tr><tr><td>1</td><td>1.1</td><td>3.0</td><td>0.9737</td></tr><tr><td>3</td><td>1.1</td><td>5.0</td><td>0.9713</td></tr><tr><td>5</td><td>1.1</td><td>7.0</td><td>0.9709</td></tr></table> Lowest Partition Entropy (PE): <table><tr><th></th><th>m</th><th>n_clusters</th><th>partition_entropy</th></tr><tr><td>1</td><td>1.1</td><td>3.0</td><td>0.0435</td></tr><tr><td>3</td><td>1.1</td><td>5.0</td><td>0.0484</td></tr><tr><td>0</td><td>1.1</td><td>2.0</td><td>0.0488</td></tr></table> Top Fuzzy Silhouette Scores: <table><tr><th></th><th>m</th><th>n_clusters</th><th>fuzzy_silhouette_score</th></tr><tr><td>19</td><td>1.5</td><td>7.0</td><td>0.5377</td></tr><tr><td>5</td><td>1.1</td><td>7.0</td><td>0.5374</td></tr><tr><td>33</td><td>2.0</td><td>7.0</td><td>0.5362</td></tr></table>		m	n_clusters	silhouette_score	20	1.5	8.0	0.4189	5	1.1	7.0	0.4180	1	1.1	3.0	0.4178		m	n_clusters	partition_coefficient	1	1.1	3.0	0.9737	3	1.1	5.0	0.9713	5	1.1	7.0	0.9709		m	n_clusters	partition_entropy	1	1.1	3.0	0.0435	3	1.1	5.0	0.0484	0	1.1	2.0	0.0488		m	n_clusters	fuzzy_silhouette_score	19	1.5	7.0	0.5377	5	1.1	7.0	0.5374	33	2.0	7.0	0.5362	The results suggest that 7 clusters consistently perform well across all metrics. Specifically: Silhouette Score peaks at 8 clusters but is nearly identical for 7, indicating strong, well-separated clusters. Partition Coefficient (PC) and Partition Entropy (PE) both favor 3 clusters, but 7 still ranks high, suggesting good fuzzy clustering quality. Fuzzy Silhouette Score is highest at 7 clusters, reinforcing that 7 is an optimal balance for structure and fuzzy membership. Overall, 7 clusters offer a strong and consistent clustering solution across both hard and fuzzy clustering metrics.
	m	n_clusters	silhouette_score																																																														
20	1.5	8.0	0.4189																																																														
5	1.1	7.0	0.4180																																																														
1	1.1	3.0	0.4178																																																														
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5	1.1	7.0	0.5374																																																														
33	2.0	7.0	0.5362																																																														

1.2 Optimal Setting Found

Algorithms: Fuzzy C Means

n_clusters	7
Fuzziness coefficient (m)	1.5
Maximum iterations (MAX_ITER)	1000
Convergence threshold (epsilon)	0.0001
Silhouette score	0.380
Fuzzy Silhouette Score	0.502

1.3 Handling Missing or Noisy Data

Following hyperparameter tuning, additional data cleaning was performed to improve model quality. Specifically, extremely frequent and rare words which can introduce noise and bias into text-based models were removed from the dataset. This was done by applying filtering rules to the article column, excluding words that were either overly common or appeared too infrequently. The goal was to reduce noise, enhance the signal in the text data and improve clustering results in the final model training phase. Below is an example of extremely common and rare words.

Extremely Common Words

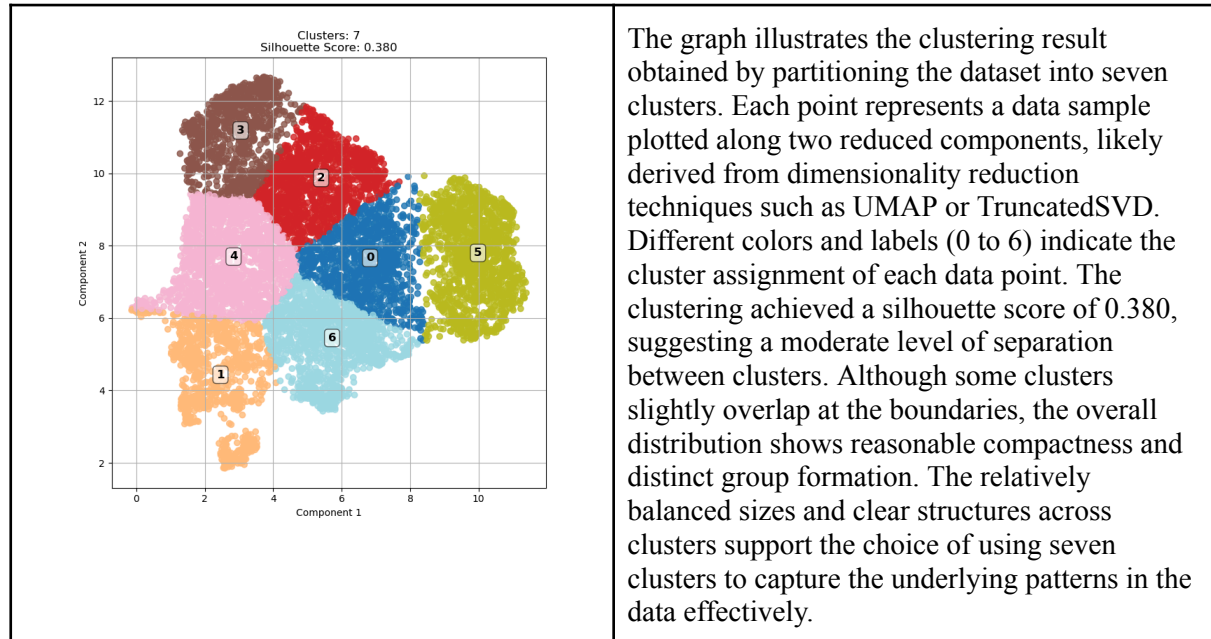
1.	say	– 53776
2.	climate	– 26127
3.	would	– 15663
4.	new	– 15089
5.	government	– 14465
6.	people	– 13680
7.	make	– 13308
8.	use	– 13218
9.	change	– 13028
10.	year	– 12868

Rare Words

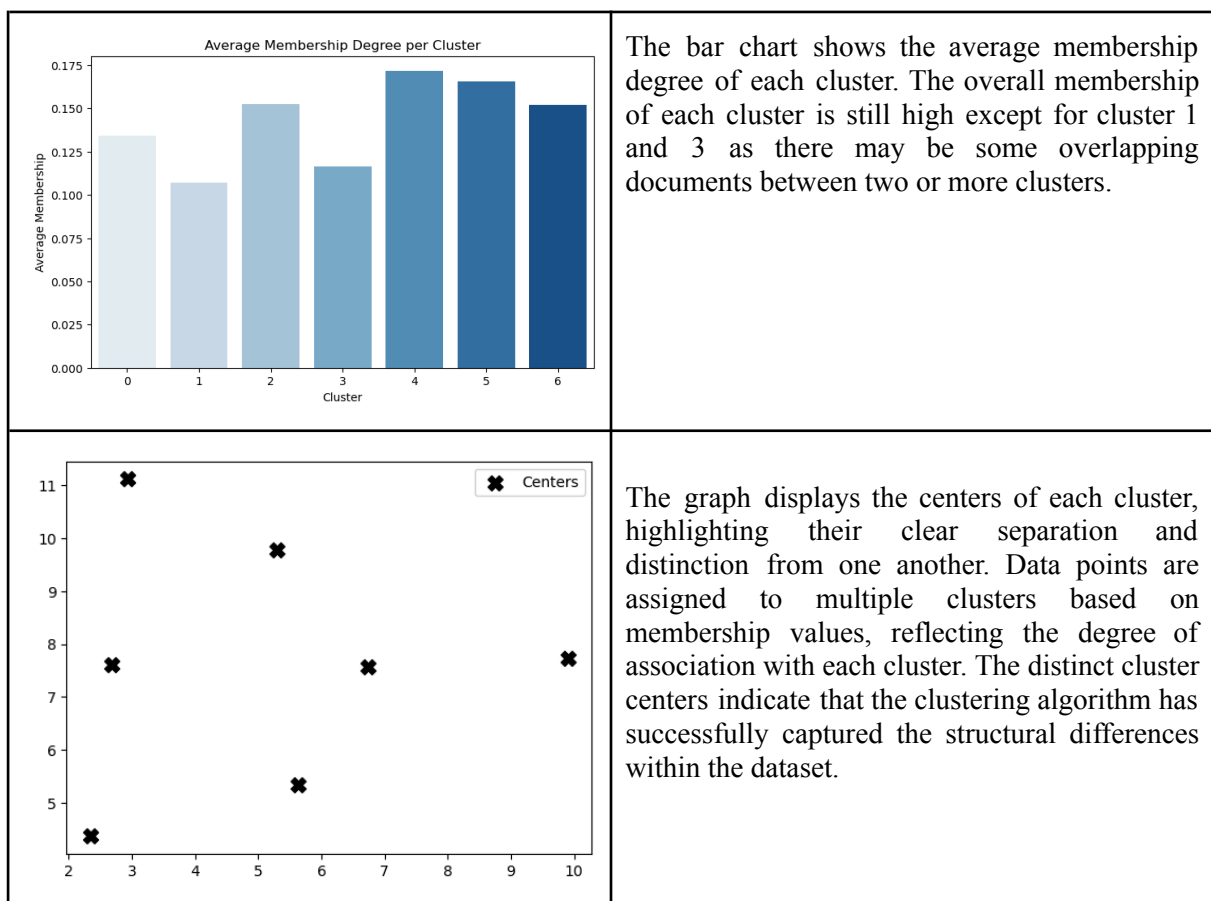
1.	rightwingers	– 9
2.	pinchbeck	– 7
3.	wakefieldread	– 2
4.	morehouseholds	– 2
5.	embroil	– 12
6.	backbenches	– 10
7.	bim	– 3
8.	afolami	– 1
9.	hitchin	– 1
10.	harpenden	– 2

2.0 Model Evaluation and Visualization

2.1 Model Evaluation



2.2 Model Visualization



3.0 Model Deployment

3.1 Deployment Tools

Streamlit was chosen for its simplicity and efficiency. It allows easy integration with Python libraries like scikit-learn, pandas and matplotlib for data processing and model predictions. Streamlit's interactive features make it easy to input data and view results in real-time. Its built-in visualization support helps display charts and graphs, while the quick setup and deployment process make it ideal for rapid prototyping and accessibility. This combination of ease of use and functionality made Streamlit the best choice for deploying the model.

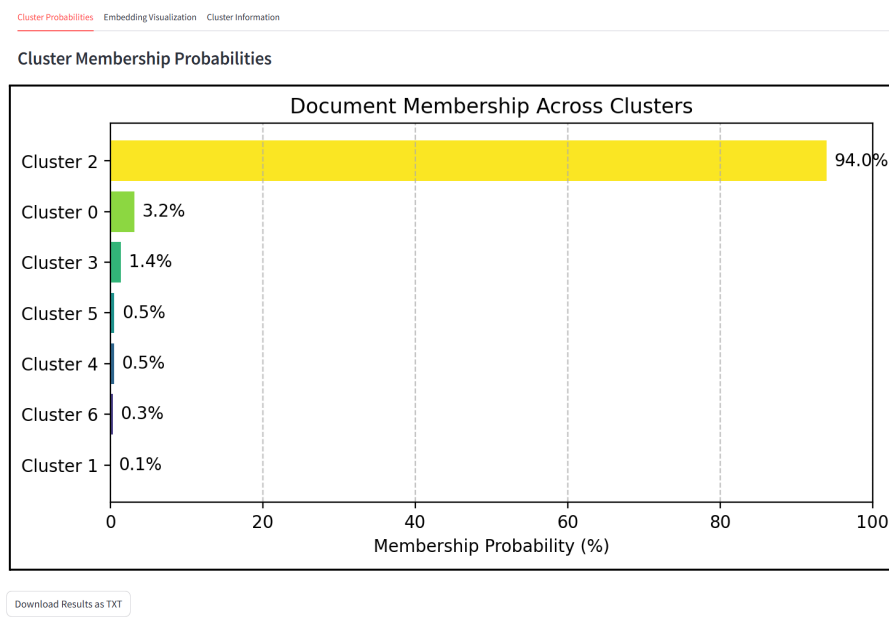
3.2 Testing on Deployed Model

Appendix 1.1 displays the main interface of the deployed model. A description bar (Appendix 1.2) provides an overview of how the model works. Users can input data either by typing or uploading a text or PDF file, as shown in Appendix 1.3. Once the "Cluster" button is clicked (Appendix 1.4), the model processes the input and performs clustering.

The results section presents several analyses. Users can view their document's primary cluster assignment along with membership scores, since Fuzzy C-Means allows soft clustering where documents may belong to multiple clusters. The processed text is also shown. The output includes a bar chart of cluster membership probabilities, a scatter plot of document embeddings highlighting the user's input, and a summary of the assigned cluster, including its main theme and key terms. This helps users quickly understand the document's dominant topic.



Figure 3.2.1 Sample of Analysis Results

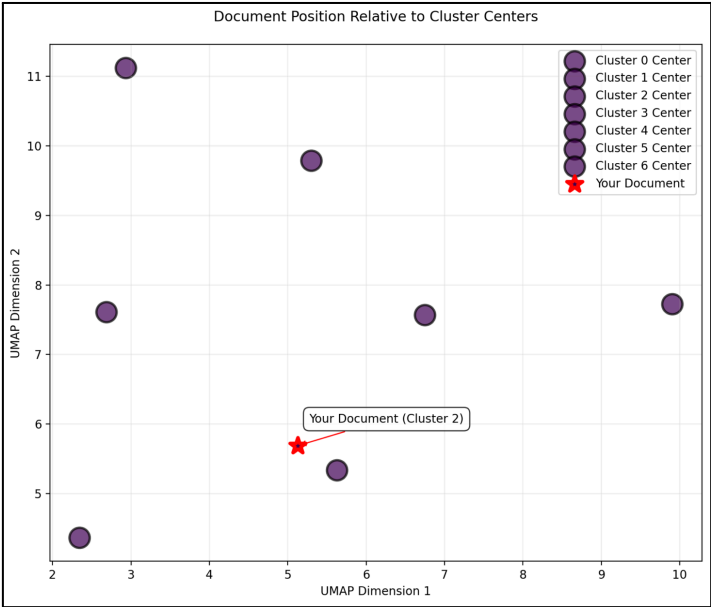


Environmental Discourse Analysis Tool | Powered by BERT and Fuzzy C-Means Clustering

For research and educational purposes

Figure 3.2.2 Cluster Membership Probabilities Results

Cluster Probabilities Embedding Visualization Cluster Information



Note: This shows the document's position in the reduced 2D space relative to the training data clusters.

[Download Results as TXT](#)

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For research and educational purposes

Figure 3.2.3 Scatter Plot of User Input across Clusters

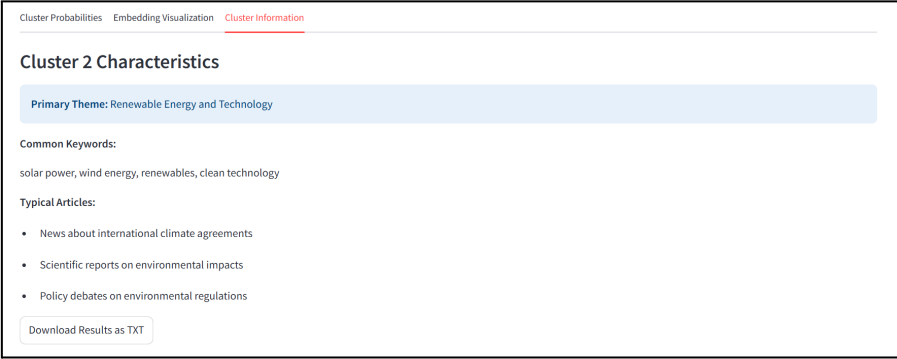


Figure 3.2.4 Cluster Information Results

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Appendix

Appendix 1.1

Environmental Discourse Clustering

Analyzing News Articles with NLP and Fuzzy Clustering

About this tool

Input Your Document

Select input method:

☒ Paste text ☐ Upload file

Enter article text here...

Paste the full text of an environmental news article...

Analyze Document

Environmental Discourse Analysis Tool | Powered by BERT and Fuzzy C-Means Clustering

For research and educational purposes

Appendix 1.2

Environmental Discourse Clustering

Analyzing News Articles with NLP and Fuzzy Clustering

About this tool

This application uses advanced Natural Language Processing (NLP) and machine learning techniques to analyze environmental news articles from The Guardian. It automatically categorizes articles into thematic clusters based on their content.

- How it works:** Upload a text document or paste article text, and the system will:
 - Process and clean the text
 - Extract semantic features using BERT embeddings
 - Reduce dimensionality with PCA and UMAP
 - Assign to a cluster using Fuzzy C-Means clustering
- Use cases:** Media analysis, content categorization, trend monitoring in environmental journalism

Appendix 1.3

Environmental Discourse Clustering

Analyzing News Articles with NLP and Fuzzy Clustering

About this tool

Input Your Document

Select input method:

☐ Paste text

☒ Upload file

Upload a text or PDF file

Drag and drop file here

Limit 200MB per file • TXT, PDF

Browse files

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Input Your Document

Select input method:

☐ Paste text

☒ Upload file

Upload a text or PDF file

Drag and drop file here

Limit 200MB per file • TXT, PDF

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ML_text.txt 0.6KB

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Processing Steps

- Text Cleaning:** Removing special characters, stopwords, and lemmatizing
- Feature Extraction:** Generating BERT embeddings (512-dimensional)
- Dimensionality Reduction:** Applying PCA and UMAP for visualization
- Cluster Assignment:** Calculating membership probabilities